

Integrative Machine Learning framework for phenotype prediction in rice

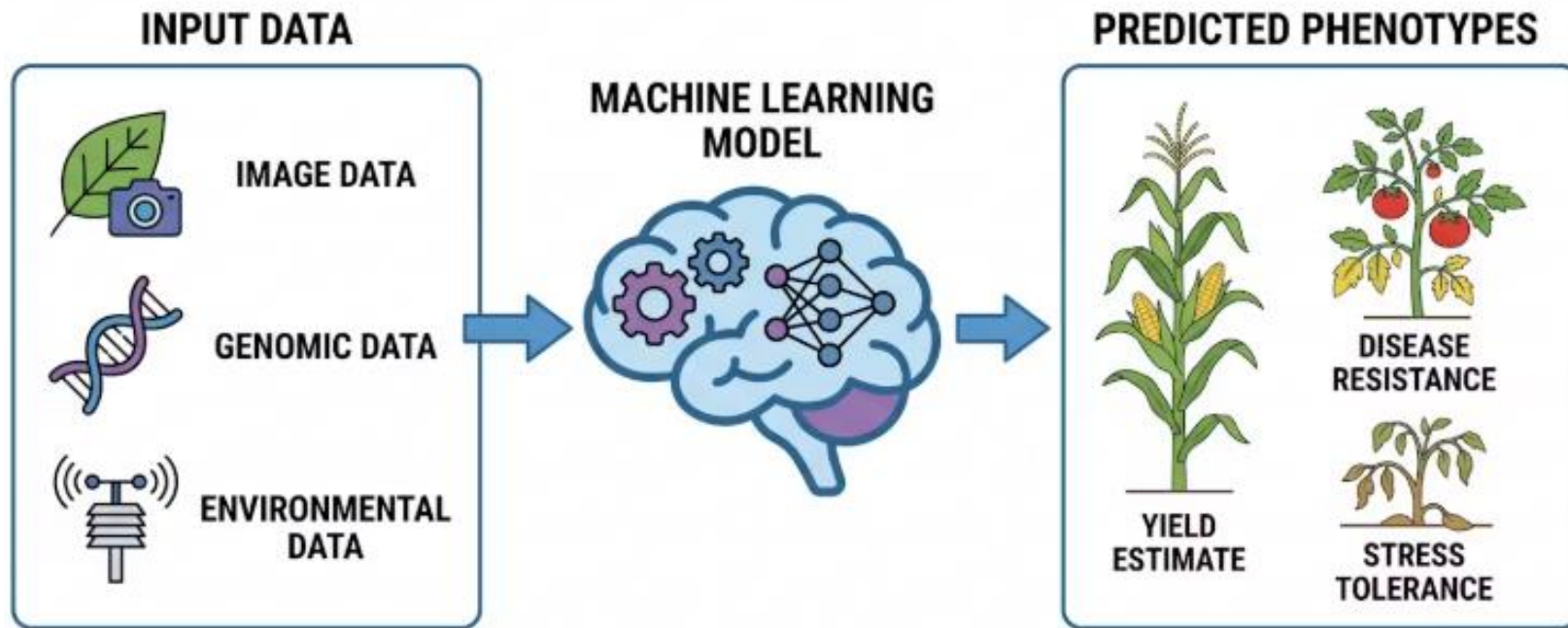
Evolution 2026

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Predicting Phenotypes: What and Why It Matters

- The Core Challenge
 - Genotype $\times$ Environment $\rightarrow$ Complex Phenotypes
- Why Phenotype Prediction Matters



Predictive Modeling

- What is Predictive Modeling ?
 - ML/DL algorithms learn **mappings** from high-dimensional features → biological outcomes
 - Goal: **predict** phenotypes, plasticity, and other traits
- Why Go Beyond Traditional Association Studies (e.g., GWAS)?
 - Integrates all features simultaneously
 - Handles correlated variables
 - Captures complex nonlinear relationships
 - Combines multiple types of data (e.g. multi-omics and environmental)

Asian Cultivated Rice



International Rice Informatics Consortium

Oryza sativa (Poaceae)

- One of the most important crops in the world
 - >20% of calories for more than half of the human population
- Two major subspecies
 - Indica (tropical)
 - Japonica (temperate)
- Rich resources available
 - ➔ Ideal system for predictive modeling

Objectives of this study

Goal: *Build an integrative ML framework to predict fitness-related phenotypes in rice*

① **Predict** fitness-related rice phenotypes

- Seed grain number (*regression*) & flowering success (*classification*)

② **Compare** modeling approaches and data tiers

- Benchmark linear, tree-based & distance-based algorithms → select the best performer
- Evaluate progressively richer data layers:
P → PG → PGT → PGTE (+GT)

③ **Quantify** data contribution & interpret biology

- Identify influential features (SHAP) → biological interpretation

④ **Test** generalizability

- Validate model on an independent external dataset (Gupta et al., 2025)

Data Collection

Dataset Overview

Dataset	Training Dataset	External Validation Dataset
Source	Groen et al., 2020	Gupta et al., 2025
Location	IRRI, Los Baños, Philippines	IRRI, Los Baños, Philippines
Year	2016 Dry Season	2017 Dry Season
Environment	Wet & Drought	Control & Salt Stress
Accessions	220 (136 indica, 84 japonica)	195 (130 indica, 65 japonica)
Individuals	1,314	1,170

Data Availability

Data Type	2016	2017	Common Features
Phenotypes	37	10	2
Genotypes (SNPs)	52,417	444,060	5,012
Gene Expression	15,632	18,141	15,394
Environmental Variables	21	NA	0

Methods



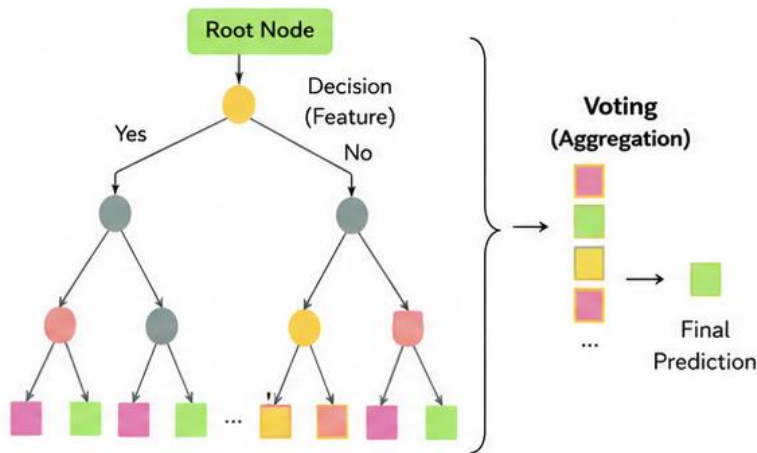
Model Training, Validation, and Evaluation Framework

- Data Splitting:
 - Training: 80% of the original data
 - Internal Validation: 20% of the original data
 - External Validation: Datasets of experiment in 2017 (Gupta et al., 2025)
- Performance Metrics:
 - Regression: R^2 $[-1, 1]$
 - How much variance the model explained
 - Classification: ROC-AUC $[0.5, 1]$
 - How well the model correctly identify TP vs. FP

Algorithm Comparison

Tree-based

Uses a tree structure of if-then rules to make predictions by splitting the data based on feature values.

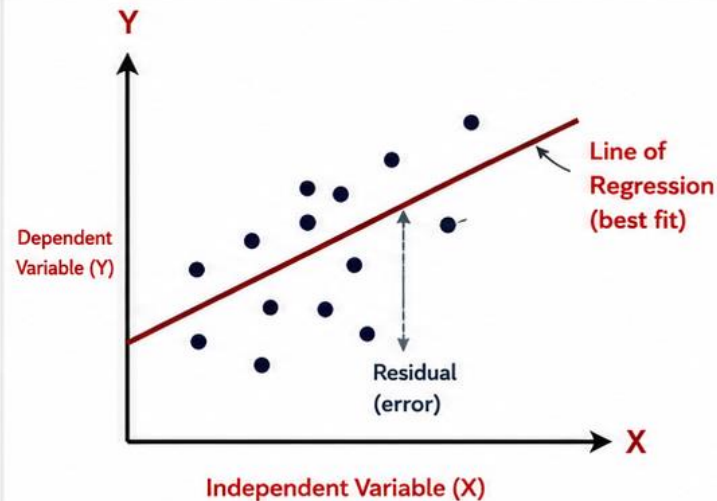


Key Idea:

Recursively split the data to create homogeneous groups and aggregate results (e.g., majority vote or average).

Linear

Assumes a linear relationship between input features and the target variable and fits a straight line (or hyperplane) to the data.

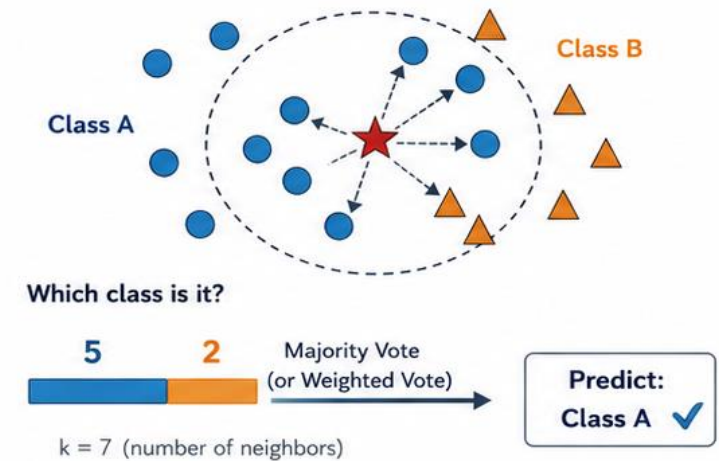


Key Idea:

Find the line (or hyperplane) that minimizes error (e.g., least squares) between predicted and actual values.

Distance-Based

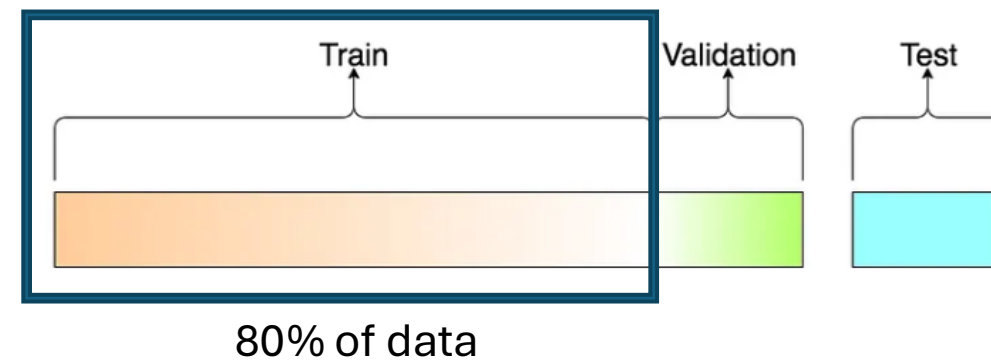
Makes predictions based on the distance (or similarity) between a new point and its neighbors in the feature space.



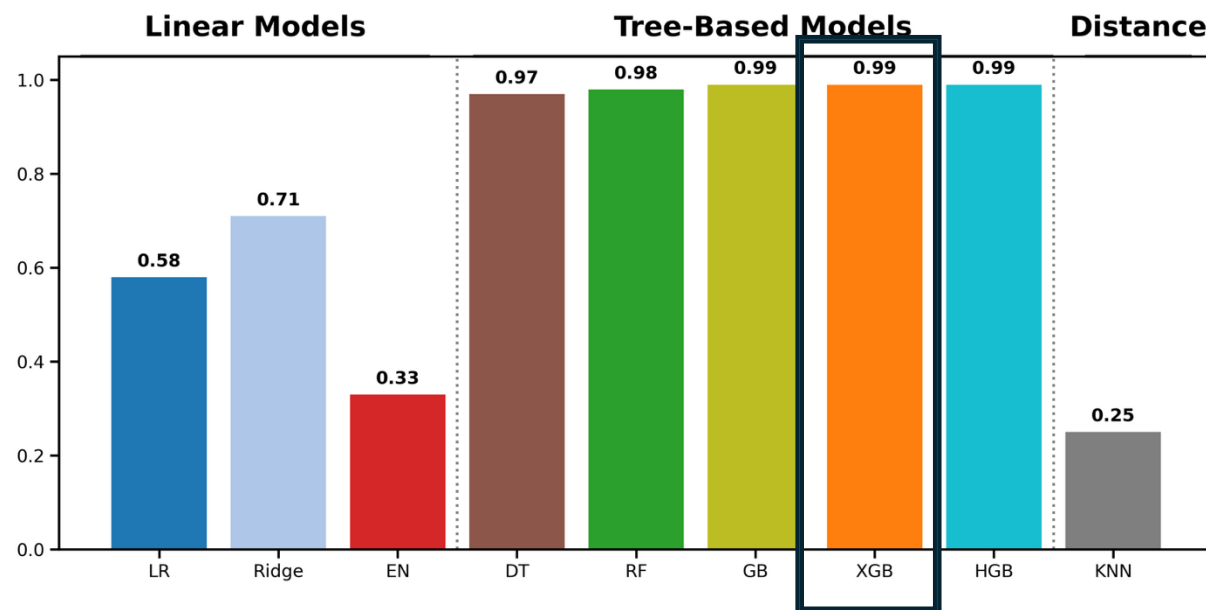
Key Idea:

Find the k nearest neighbors and predict based on the majority (or weighted) vote or average of their values.

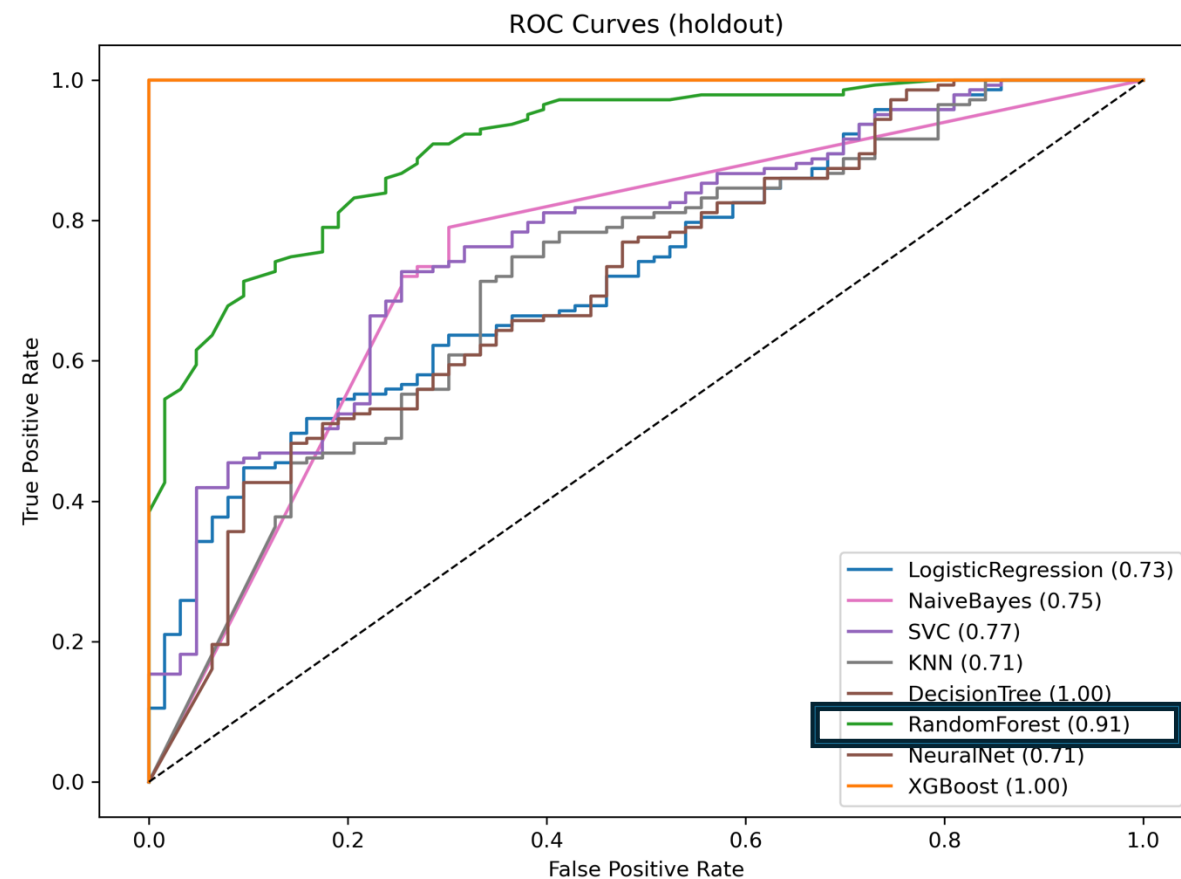
Results 1 - Training Evaluation (PGTE)



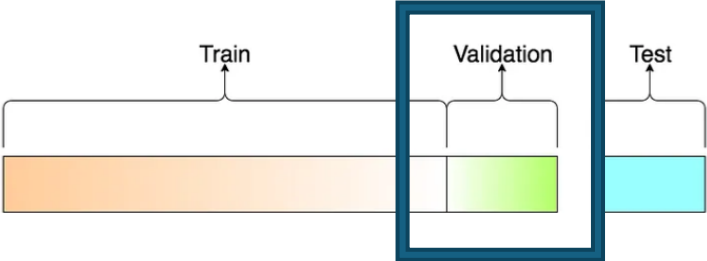
Regression Models



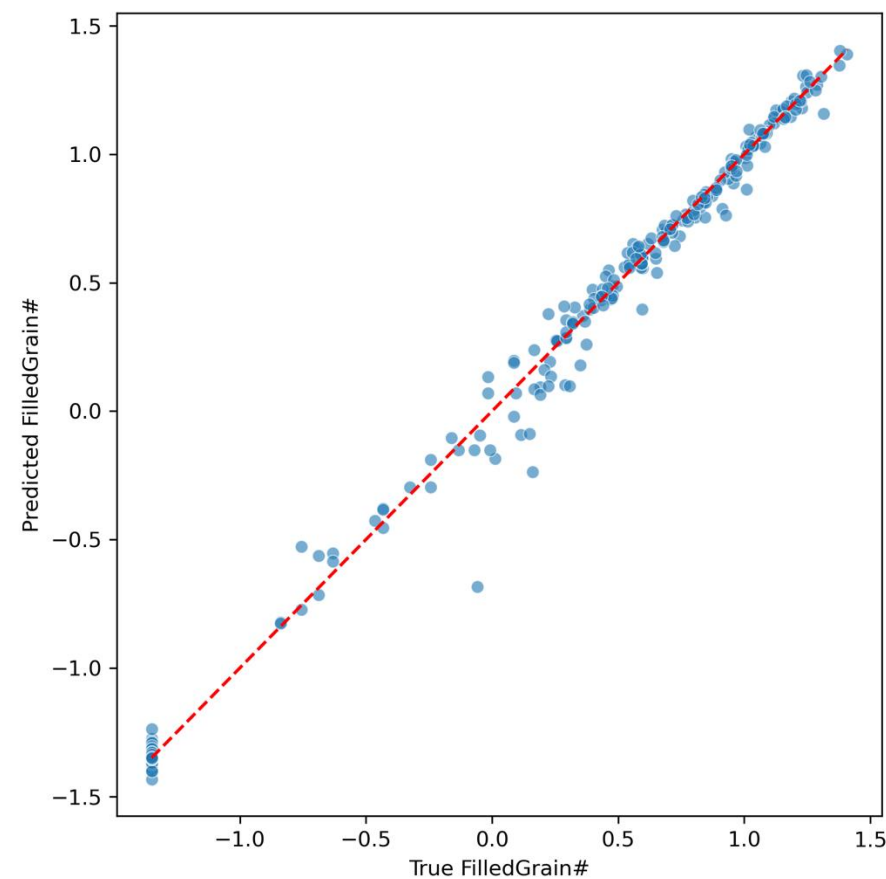
Classification Models



Results 2 - Internal Validation (PGTE)

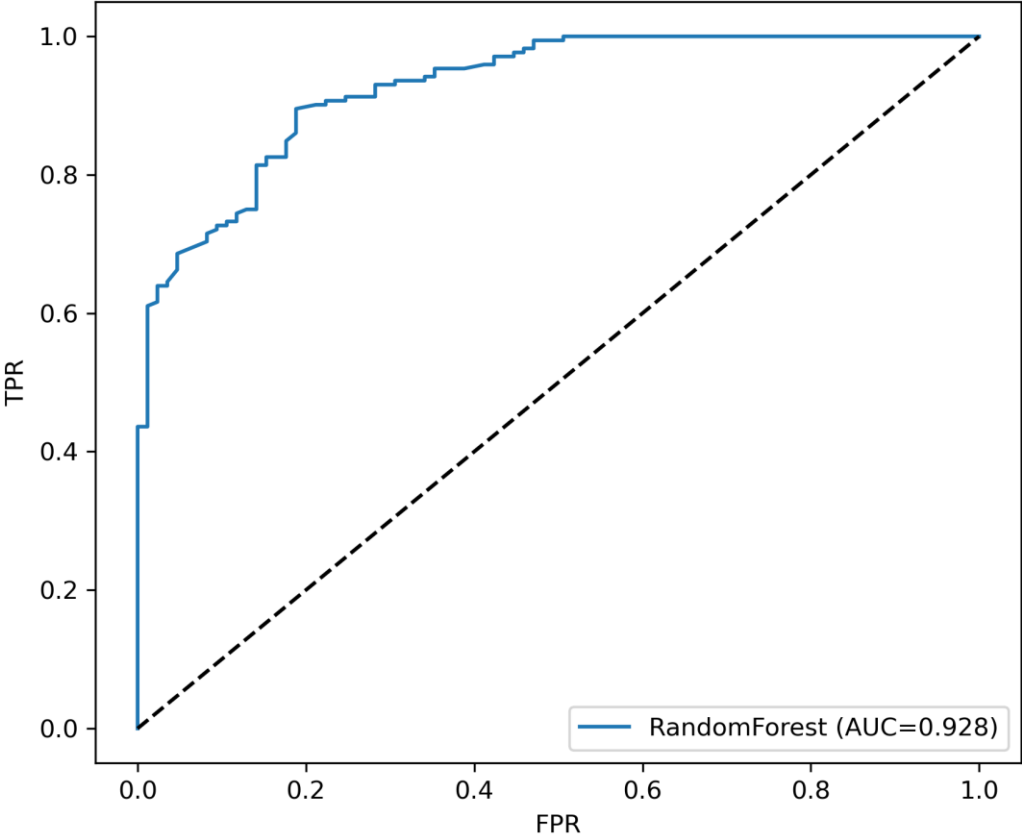


Regression Model (XGB)



$r^2 = 0.98$ (XGB)

Classification Model (RF)

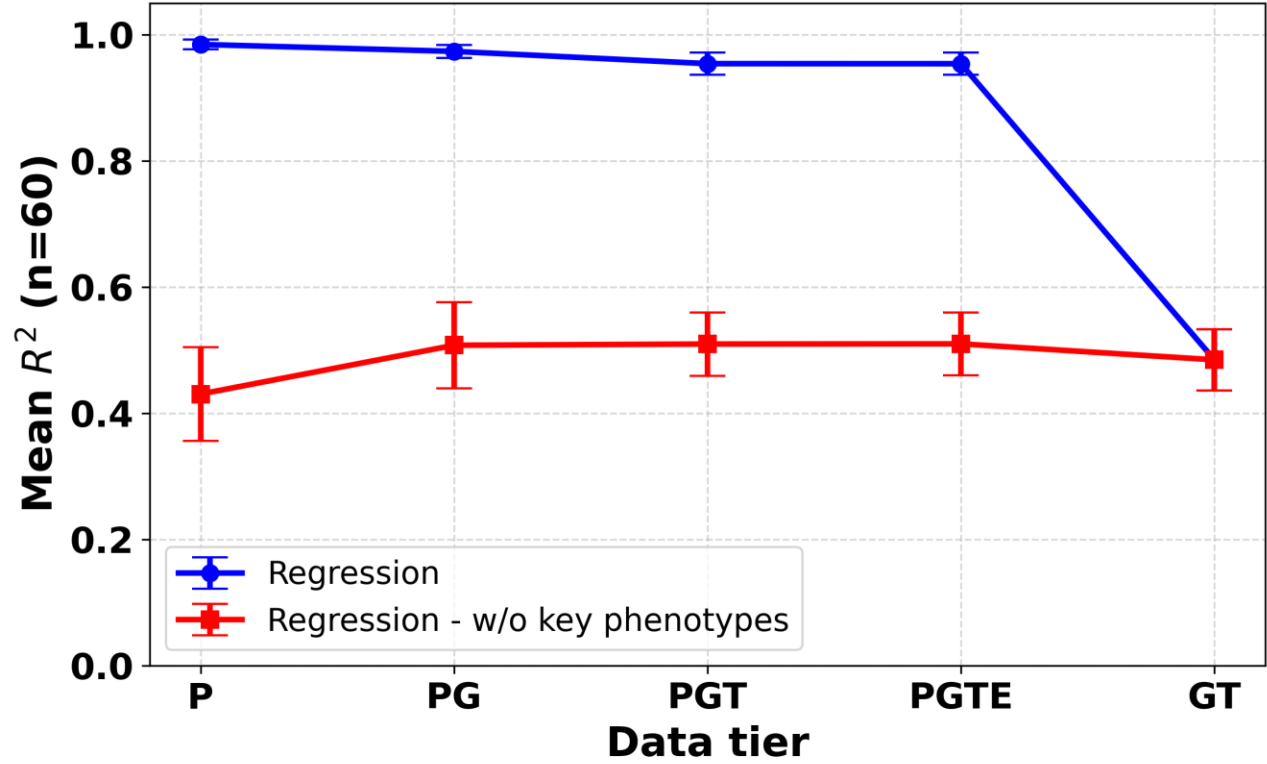


ROC-AUC = 0.93 (RF)

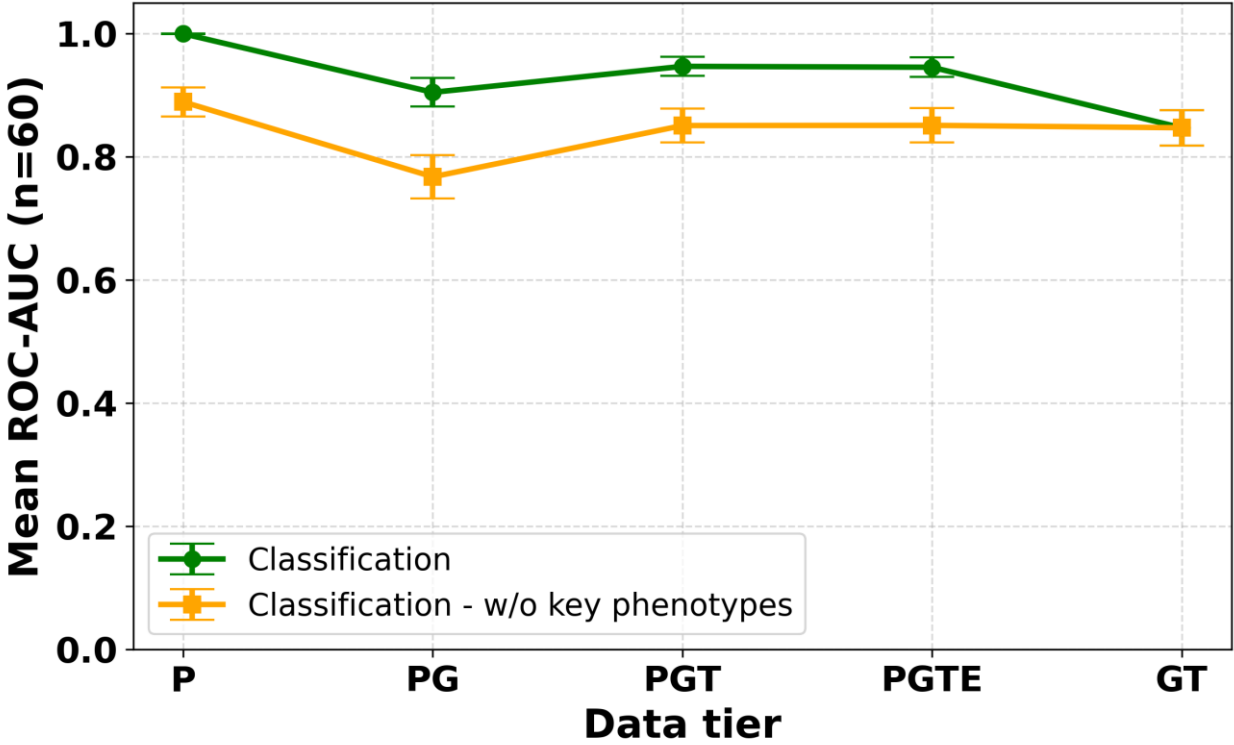
20% of data

Results 3 - Prediction Model Performances per Data Tier

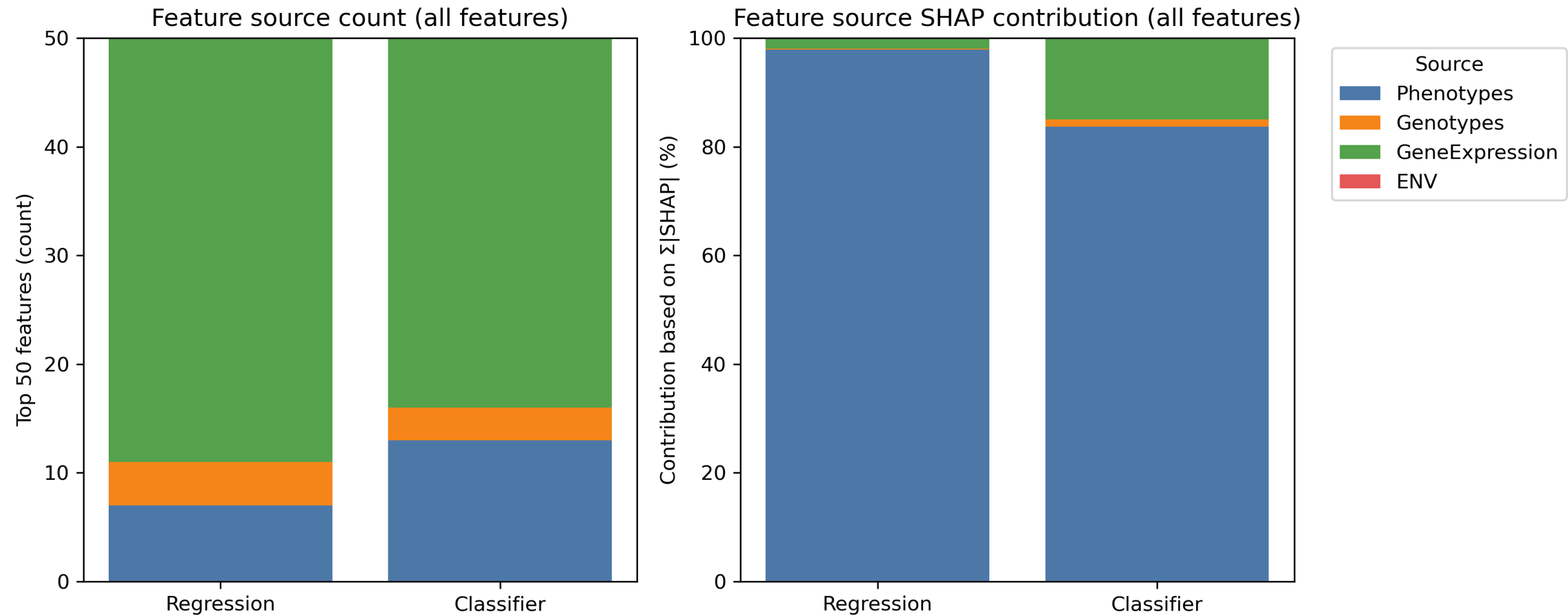
Regression Model (XGB)



Classification Model (RF)



Feature Importance Analysis (PGTE)



Biological Interpretation of Key Predictors (PGTE)

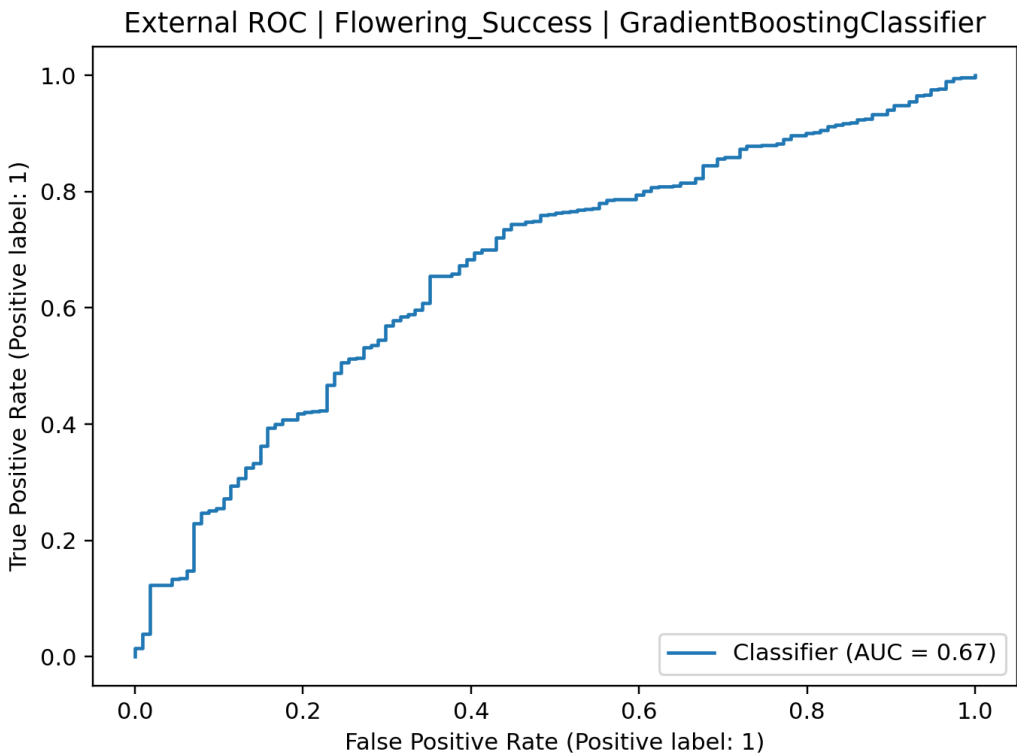
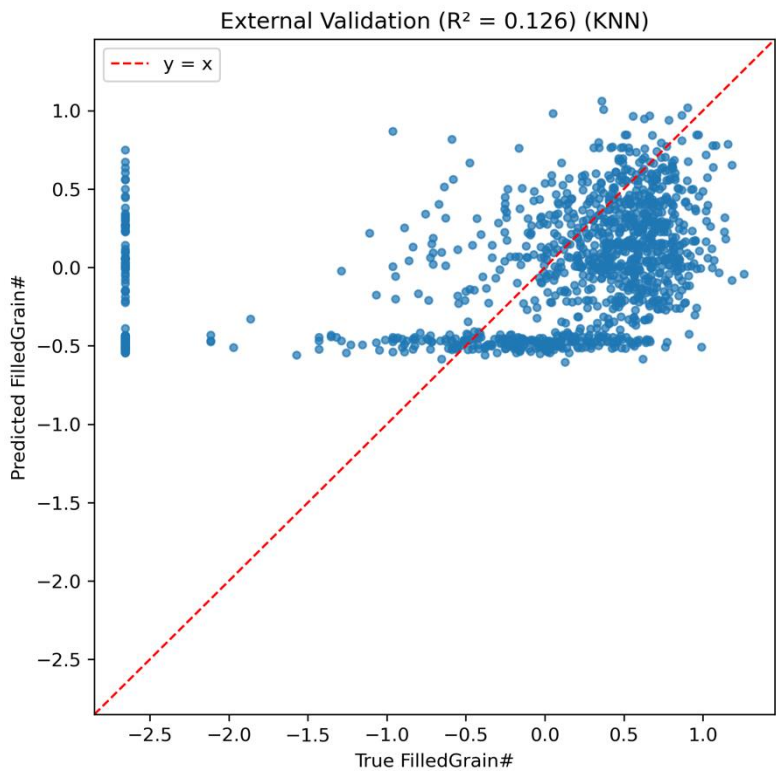
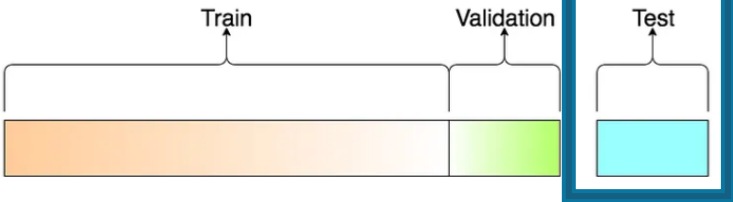
Filled Grain Numbers

- **Grain production**
 1. Spikelet fertility (#1) — direct determinant of filled grain production
 2. Spikelet# (#2) — defines reproductive sink size and yield capacity
- **Flowering / drought escape**
 1. DTF 50% (#12) — flowering time affects drought escape and grain filling
- **Stress-response**
 1. SNP 2_25864639 (#4) — linked to stress-response locus
- **Metabolism / enzyme function**
 1. OsMIOX (#6) — myo-inositol metabolism; osmotic adjustment
 2. OsSSIIIb (#7) — starch biosynthesis and seed filling

Flowering Success

- **Reproductive success**
 1. DTF 50% (#12) — timing of flowering influences successful reproduction
 2. RPS20 (#9) — translation and cellular growth during reproduction
 3. PPR10 (#10) — organellar gene expression and reproductive development
- **Stress-response / adaptation**
 1. PUB68 (#8) — ubiquitin-mediated stress signaling
 2. OsLTPG12 (#15) — lipid transfer, membrane function, stress tolerance
 3. ALBA6 (#22) — DNA/RNA binding; drought, heat, cold, and salt tolerance
- **Metabolism**
 1. ALPHACA7 (#20) — carbonic anhydrase; photosynthesis and reproductive performance

External Validation (GT)



Model	Algorithm	Target	R2/ROC	N_train	N_val	Features
Regression	KNN	FilledGrain#	0.13	1314	1170	20406
Classification	GB	Flowering Success	0.67	1314	1170	20406
Classification	RF	Subspecies	0.98	1314	1170	20406

Conclusion

- Fitness-related phenotypes were highly predictable (under training and interval)
 - ($R^2 = 0.98$ ROC-AUC = 0.93)
- Tree-based models consistently outperformed linear and distance-based approaches (except external validation of regression model)
- Phenotypes contributed most to prediction accuracy
- SHAP analysis identified key fitness-related predictors
- External validation demonstrated partial generalizability

Discussion & Future Directions

- Why there was huge discrepancy between training and ext-validation?
- Biological Factors
 1. Different stress environments
 - Training (drought) vs. External Validation (salt stress) → alter genotype-phenotype relationships
 2. Transcriptomic differences
 - environment-dependent gene expression likely reduced transferability.
 3. Genotypic marker discrepancy (LD-pruned vs. raw)
 - ↑ SNPs
- Methodological Factors
 1. Potential accession-level bias
 - ~1,300 plants from 220 accessions possibly inflated internal validation performances
 2. Unoptimized feature selection and parameters
- Future Directions
 1. Develop multi-modal ML/DL approaches to increase transferability
 2. Consilience with association studies

Presenter

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Mentor

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NSF IOS 2204374

Franks Lab !



Github

Reference

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